

Q.1 FULL JOIN.

Query:

```
SELECT  
  
S1.SUB_ID,  
  
S1.SUB_NAME,  
  
C1.ROOM_NAME  
  
FROM SUBJECT S1  
  
LEFT JOIN CLASSROOM C1 ON S1.SUB_ID=C1.SUB_ID  
  
UNION  
  
SELECT C1.CLASS_ID,  
  
C1.ROOM_NAME,  
  
S1.SUB_ID  
  
FROM CLASSROOM C1  
  
RIGHT JOIN SUBJECT S1 ON  
  
C1.SUB_ID=S1.SUB_ID;
```

Screenshot:

The screenshot displays the MySQL Workbench interface. The main window shows a SQL query in the editor, which is a UNION of two SELECT statements. The first SELECT statement joins SUBJECT S1 and CLASSROOM C1 on S1.SUB_ID=C1.SUB_ID, selecting S1.SUB_ID, S1.SUB_NAME, and C1.ROOM_NAME. The second SELECT statement selects C1.CLASS_ID, C1.ROOM_NAME, and S1.SUB_ID from CLASSROOM C1, right-joining it to SUBJECT S1 on C1.SUB_ID=S1.SUB_ID. The query is executed, and the results are shown in the Result Grid. The Result Grid displays two tables: the first table has columns SUB_ID, SUB_NAME, and ROOM_NAME, and the second table has columns CLASS_ID, ROOM_NAME, and SUB_ID. The bottom panel shows the Output window with a message indicating that the query was completed successfully.

MySQL Workbench

Local instance MySQL80

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Navigator

SCHEMAS

Filter objects

Tables

Views

Stored Procedures

Functions

sys

SQL3 SQL3 SQL3 SQL3 DB3 DB3 SQL File 26* SQL File 27* Q5 Q5* DB3* SQL3 Q2*

Limit to 1000 rows

154 SELECT
155 S1.SUB_ID,
156 S1.SUB_NAME,
157 C1.ROOM_NAME
158 FROM SUBJECT S1
159 LEFT JOIN CLASSROOM C1 ON S1.SUB_ID=C1.SUB_ID
160
161 UNION
162 SELECT C1.CLASS_ID,
163 C1.ROOM_NAME,
164 S1.SUB_ID
165 FROM CLASSROOM C1
166 RIGHT JOIN SUBJECT S1 ON
167 C1.SUB_ID=S1.SUB_ID;

Result Grid

Filter Rows

Exports

Wrap Cell Contents

SUB_ID	SUB_NAME	ROOM_NAME
CS101	Introduction to Database	101-A
CS102	Artificial Intelligence	102-B
HI101	Ancient Civilizations	101-B
MA201	Linear Algebra	201
PH301	Quantum Mechanics	Lab-3

Result 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	20:39:17	SELECT C1.ROOM_NAME, S1.SUB_NAME, S1.SUB_ID FROM SUBJECT S1 LEFT JOIN CLASSROOM C1...	10 row(s) returned	0.000 sec / 0.000 sec
16	20:39:49	SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME S1.SUB_NAME S1.SUB_ID FROM SUBJECT S1 ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
17	20:39:53	SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME S1.SUB_NAME S1.SUB_ID FROM SUBJECT S1 ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
18	20:39:59	SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME, S1.SUB_NAME, S1.SUB_ID FROM SUBJECT S1 ...	Error Code: 1222. The used SELECT statements have a different number of columns	0.000 sec
19	20:40:08	SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME, S1.SUB_ID FROM SUBJECT S1 LEFT JOIN CLA...	Error Code: 1222. The used SELECT statements have a different number of columns	0.016 sec
20	20:40:15	SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME, FROM SUBJECT S1 LEFT JOIN CLASSROOM C...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
21	20:40:21	SELECT S1.SUB_ID, S1.SUB_NAME, C1.ROOM_NAME, FROM SUBJECT S1 LEFT JOIN CLASSROOM C...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec / 0.000 sec

Query Completed

25°C
Light rain

20:49
01-07-2028

Insights:

- Room name for HI101 is empty since the classroom table doesn't contain the HI101 as a value in SUB_ID.
- Though SUB_ID HI101 is fetched, as per the union operator's function.
- As it fetches the unique data removing duplications and combines the rows of both the tables.